

FINALIZED CARE PLAN

PATIENT NAME

Homo Sapiens (Female,
Asthma & Respiratory, 42y)

VISIT DATE

September 30, 2026

PREPARED BY

Pocket Gull AI

BP

118/72

HR

76 bpm

TEMP

98.2°F

SPO2

99%

WEIGHT

145 lbs

HISTORICAL CONDITIONS

Asthma

Anxiety Disorder

Chronic Lower Back Pain

Opioid Use Disorder (in remission)

Depression



ACTIVE CARE PLAN

CLINICAL ASSESSMENT

Sarah Jenkins is a 42-year-old female presenting with chronic lower back pain (L4-L5 radiculopathy), opioid use disorder in sustained remission, and co-morbid depression. The primary clinical challenge is multi-modal pain management that avoids opioid relapse risk while addressing the functional and psychological dimensions of her condition.

PRIORITY LIST

- Opioid Relapse Prevention: Highest-risk factor; pain management strategy must explicitly exclude opioid analgesics.
- Chronic L4-L5 Radiculopathy: Constant aching with left gluteal radiation, pain level 7/10. Worsens with prolonged standing.
- Co-morbid Depression: Active depression compounds pain tolerance and functional recovery; concurrent psychiatric management is essential.
- Functional Deconditioning: Pain-avoidance behavior likely contributing to progressive deconditioning.

PLAN OF CARE

- Immediate: Refer to a pain management specialist with expertise in non-opioid modalities. Initiate NSAID trial (e.g., naproxen 500mg BID with food) if no contraindications.
- Short-Term: Enroll in a structured physical therapy program focused on lumbar stabilization and McKenzie method exercises. Initiate CBT-based pain management (8-12 sessions).
- Ongoing: Maintain psychiatric follow-up for depression management. Consider duloxetine (SNRI) as it addresses both depression and neuropathic pain.

INTERVENTION	TARGET	FREQUENCY
Physical Therapy	Lumbar stabilization	2-3x/week x 8 weeks
CBT (Pain-focused)	Catastrophizing, coping	Weekly x 12 weeks
Duloxetine (discuss)	Depression + pain	Daily (psych-guided)
NSAID (naproxen)	Acute pain relief	BID with food, reassess at 4 weeks

GOALS

- Short-term (2 weeks): Pain VAS score reduction from 7/10 to <=5/10. PT program initiated.
- Long-term (3 months): Functional ADL capacity restored to baseline. Sustained opioid abstinence maintained. PHQ-9 score reduced by >=5 points.

REFERENCES

- Dowell, D. et al. (2022). CDC Clinical Practice Guideline for Prescribing Opioids for Pain. MMWR, 71(3), 1-95. DOI: [10.15585/mmwr.rr7103a1](https://doi.org/10.15585/mmwr.rr7103a1). Peer-Reviewed.

DIAGNOSTIC WORKUP

TEST	RATIONALE	PRIORITY
Updated MRI lumbar spine	Confirm L4-L5 disc status, nerve root compression	High
PHQ-9 (depression screen)	Baseline severity score for treatment tracking	High
Basic metabolic panel	Pre-NSAID renal/hepatic safety check	Medium
DXA bone scan	Rule out osteoporosis contributing to pain	Low

NUTRITION & LIFESTYLE

- **Anti-inflammatory diet:** Increase omega-3 fatty acid intake (oily fish, flaxseed). Reduce ultra-processed foods and refined sugars, which are associated with heightened systemic inflammation [Calder et al., 2017].
- **Maintain healthy weight:** Current BMI within normal range; avoid weight gain as it increases lumbar load.
- **Sleep hygiene:** Depression and pain both disrupt sleep architecture. Establish consistent sleep/wake schedule; limit screen time 1 hour before bed.
- **Alcohol avoidance:** Given OUD history, minimize all substance use including alcohol (CNS depressant with low relapse threshold overlap).
- **Standing workstation or ergonomic assessment:** Reduce time in provocative posture at work.

SUPPLEMENTATION

SUPPLEMENT	DOSE	FREQUENCY	RATIONALE
Magnesium glycinate	300-400mg	Nightly	Muscle relaxation, sleep support
Vitamin D3 + K2	2000 IU D3 / 100mcg K2	Daily with fat	Bone health, pain modulation
Omega-3 (EPA/DHA)	2g EPA+DHA	Daily with meal	Anti-inflammatory support

MOVEMENT & REHABILITATION

- **McKenzie Method (extension-biased exercises):** Highly effective for L4-L5 disc pathology with peripheral symptoms. Begin with supervised sessions; transition to home program.
- **Core stabilization (McGill Big Three):** Bird-dog, curl-up, side plank. Engage deep spinal stabilizers without compressive loading.
- **Aquatherapy:** Low-impact option for acute flare periods; reduces joint load while maintaining cardiovascular function.
- **Avoid:** High-impact activities, heavy deadlifts, and prolonged flexion-dominant postures until PT-cleared.

MIND-BODY

- **Mindfulness-Based Stress Reduction (MBSR):** 8-week structured program shown to reduce pain catastrophizing and improve depression scores in chronic pain populations.
- **Diaphragmatic breathing:** Activates parasympathetic nervous system, reducing pain perception threshold.

BIOCHEMICAL ASSESSMENT

The patient displays elevated systemic inflammatory markers and increased oxidative stress. Suboptimal levels of key micronutrients are present, specifically in vitamin D3 and magnesium, which limits cellular energy efficiency.

NUTRITION FOCUS

- Anti-inflammatory pathways: High oxidative stress requires active anti-inflammatory dietary inputs.
- Mitochondrial cofactor support: Cellular respiration needs magnesium and B-vitamins for efficient ATP production.
- Bone and nerve support: Suboptimal vitamin D3 needs tailored nutritional support.

NUTRITIONAL INTERVENTIONS

NUTRIENT/COMPOUND	THERAPEUTIC DOSE	DELIVERY METHOD	PRIMARY PATHWAY
Vitamin D3	2000 IU	Oral with fat	Bone health and immune support
Magnesium glycinate	400 mg	Oral nightly	NMDA regulation & muscle relaxation
Omega-3 (EPA/DHA)	2g	Oral with meals	Systemic inflammation reduction

DIETARY ADJUSTMENTS

- Mediterranean Diet Pattern: High consumption of olive oil, leafy greens, and nuts to lower inflammatory cytokines.
- Gluten-Free Trial: Consider 4-week trial to reduce systemic immune reactivity and intestinal permeability.
- Increased Hydration: Minimum 2.5L filtered water daily to assist in renal clearance of metabolic waste.

IMMEDIATE (24-72 HOURS)

- Confirm PT referral appointment scheduled within 1 week.
- Obtain baseline PHQ-9 depression score and document in chart.
- Verify no active opioid prescriptions from other providers (prescription drug monitoring check).
- Review NSAID for GI contraindications; prescribe PPI co-therapy if indicated.

SHORT-TERM (WEEK 1-2)

- PT intake session completed; home exercise program assigned.

- First CBT session scheduled.
- Follow-up call at Week 2 to assess NSAID tolerability and pain VAS score.
- Psychiatry/prescriber consultation for duloxetine initiation if patient amenable.

ONGOING (MONTH 1-3)

PARAMETER	TARGET	FREQUENCY	ESCALATION TRIGGER
Pain VAS	<=5/10	Monthly	Pain escalation >8/10 or new neurological symptoms
PHQ-9 Score	Reduction >=5 pts	Monthly	Score >15 or suicidal ideation -> urgent referral
PT Adherence	>=80% sessions attended	Bi-weekly check	<50% adherence -> reassess barriers
Opioid Abstinence	Sustained remission	Each visit	Any opioid use -> addiction medicine consult
Functional ADLs	Able to stand >20 min	Monthly	No improvement -> pain specialist referral

RED FLAGS

WARNING: Seek Immediate Clinical Attention if:

- New or worsening bilateral leg weakness or numbness
- Bowel or bladder dysfunction (cauda equina syndrome)
- Pain VAS 9-10/10 unresponsive to current regimen
- Patient reports opioid cravings or use
- PHQ-9 >=20 or active suicidal ideation

UNDERSTANDING YOUR PAIN & RECOVERY PLAN

Your lower back pain is real and challenging, but there are effective ways to manage it without returning to opioid medications. Your recovery relies on a multi-pronged approach that combines targeted physical therapy, non-opioid medications, and emotional support.

KEY POINTS TO REMEMBER

- **Opioid Safety:** Returning to opioids presents a significant health risk. Your pain management plan is designed to be completely opioid-free.

- **Stay Active:** Gentle, consistent movement through physical therapy will help strengthen your back muscles and protect your lumbar spine. Rest when needed, but avoid prolonged bed rest.
- **Mental Health Matters:** Depression and pain affect the same neural pathways in the brain. Treating your depression with your provider's help will directly improve your ability to cope with pain.

YOUR DAILY ROUTINE

- **Morning:** Take prescribed non-opioid pain medication with breakfast. Perform 10-15 minutes of prescribed home physical therapy exercises.
- **Afternoon:** Take short walking breaks every 1-2 hours if working at a desk. Apply ice or moist heat to your lower back for 15-20 minutes as needed.
- **Evening:** Practice relaxation or mindfulness techniques before bed to help manage both pain and stress. Take nightly supplements as recommended.

WHEN TO CONTACT YOUR PROVIDER

- Pain suddenly worsens or spreads further down your leg.
- You experience new weakness, numbness, or tingling in your legs or feet.
- You experience any change in bowel or bladder function (seek immediate emergency care).
- You feel overwhelming stress or urges to misuse substances.

PHYSIONET 2026 DIGITAL SIGNAL & ELECTROPHYSIOLOGY SUMMARY

Integrated EDF/PhysioNet waveform metrics captured from high-frequency BLE telemetry sensors and continuous multi-lead ECG monitoring.

ELECTROCARDIOGRAPHIC WAVEFORM MORPHOLOGY

- **QRS Interval Duration:** 92 ms (Normal baseline: < 120 ms). Zero bundle branch block patterns observed.
- **ST-Segment Elevation/Depression:** Neutral ST morphology across Lead II, V2, V5 (+0.04 mV deviation). Low acute ischemic risk.
- **QTc Interval (Fridericia):** 418 ms (Normal baseline: < 450 ms). Safe threshold maintained for concurrent SNRI / psychotropic management.

HEART RATE VARIABILITY (HRV) SPECTRAL POWER DENSITY

PARAMETER	VALUE	REFERENCE RANGE	INTERPRETATION
LF Power (0.04-0.15 Hz)	520 ms ²	300-800 ms ²	Sympathetic / Baroreflex Modulation

PARAMETER	VALUE	REFERENCE RANGE	INTERPRETATION
HF Power (0.15-0.40 Hz)	480 ms ²	200-600 ms ²	Parasympathetic / Vagal Tone
LF/HF Spectral Ratio	1.08	0.8 - 1.5	Balanced Autonomic Nervous System Homeostasis
SDNN (Time Domain)	58 ms	> 50 ms	Optimal Autonomic Recovery Buffer

CLINICAL ACTIONABLE FINDINGS

- Autonomic Tone: Balanced LF/HF ratio (1.08) indicates satisfactory stress resilience with zero autonomic crash signals.
- Arrhythmia Screening: Zero premature ventricular contractions (PVCs) or atrial fibrillation episodes detected over 24-hour continuous stream.
- Follow-up Protocol: Continue 24-hour continuous BLE telemetry monitoring during physical therapy stress trials.

BIOCHEMICAL & BIOMARKER MATRIX

The patient exhibits moderate systemic stress combined with suboptimal cellular energy production cofactors. Methylation pathway efficiency is reduced, contributing to neuroinflammation and pain sensitization. Primary focus is to support methylation pathways, improve mitochondrial respiration, and modulate NMDA receptor excitability.

```
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  { "name": "Zinc", "level": "Sub-optimal", "pathway": "Immune / Hormones" }
]
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DETECTED DEFICIENCIES

- Magnesium (Sub-optimal): Consistent with chronic pain sensitization. Magnesium blocks NMDA receptors and limits central sensitization; deficiency amplifies pain perception.
- Vitamin D3 (Sub-optimal): Reduced levels associated with increased musculoskeletal pain, depression severity, and immune dysregulation.
- Zinc (Sub-optimal): Zinc depletion impairs immune surveillance, slows tissue repair, and may worsen depression through altered dopaminergic signaling.
- Potential Folate/B9: Chronic NSAID use depletes folate; assess homocysteine for methylation cycle blockage.

ORTHOMOLECULAR PROTOCOL

INTERVENTION/MOLECULE	THERAPEUTIC DOSE	DELIVERY METHOD	TARGETED PATHWAY
Magnesium Glycinate	400mg elemental Mg	Oral, nightly with food	NMDA antagonism / ATP Synthesis / Sleep
Vitamin D3 + K2	5000 IU D3 / 100mcg MK-7	Oral with fat-containing meal	Bone mineral density / Immune modulation
Zinc Picolinate	30mg	Oral with food (avoid with calcium)	Immune function / Hormonal balance
Methylcobalamin (B12)	1000mcg	Sublingual (if GI absorption uncertain)	Myelin sheath integrity / Methylation
P-5-P (Active B6)	50mg	Oral with food	Neurotransmitter synthesis / Homocysteine

CAUTIONS & INTERACTIONS

- **Magnesium + NSAIDs:** Concurrent use is generally safe; however, high-dose NSAIDs reduce magnesium absorption — monitor renal function.
- **Vitamin D3 Toxicity Threshold:** Doses >10,000 IU/day long-term carry hypercalcemia risk; recheck 25(OH)D at 3 months.
- **Zinc + Copper Antagonism:** Doses >40mg zinc/day require copper co-supplementation (1–2mg/day) to prevent induced copper deficiency.
- **B6 (P-5-P) Neuropathy:** Doses >200mg/day of P-5-P can paradoxically induce peripheral neuropathy; stay at therapeutic 50mg dose.

CARE PLAN INSTRUCTIONS

- ☐ [Goal]: Establish asthma action plan, consider biologic dupilumab initiation, and deploy HEPA environmental filtration.

OXIDATIVE STRESS MARKERS

- Malondialdehyde (MDA): 2.4 µmol/L
- F2-Isoprostanes: 38 pg/mg creatinine

ANTIOXIDANT SOURCES

- Vitamin C: 1.2 mg/dL
- Glutathione (GSH): 1.8 µmol/g Hb

 MEDICATIONS

- Albuterol Inhaler: 2 puffs PRN
- Sertraline: 50mg Daily
- Gabapentin: 300mg TID
- Ibuprofen: 600mg PRN MSK Pain

CLINICIAN SIGNATURE

DATE & TIME

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This care plan was generated with AI assistance. Review and approval by a licensed clinician is required before implementation.

`window.addEventListener('DOMContentLoaded', () => { setTimeout(() => { window.print(); }, 400); });`